

Securing Federated TinyML Intrusion Detection for IoT/MQTT Networks with ASCON-128 in an Edge-Fog-Cloud Architecture

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Abstract. Intrusion detection in resource-constrained IoT/MQTT networks must simultaneously address three goals that the literature usually studies in isolation: low-footprint inference on microcontrollers, privacy-preserving distributed training, and authenticated communication of model artifacts. We present an end-to-end Edge-Fog-Cloud system that integrates the three on real hardware. An ESP32-S3 executes a compact 1,139-parameter TinyML classifier (≈ 4.5 kB) over thirteen flow features, while a Raspberry Pi 4 gateway performs local training and a PC server coordinates hierarchical federated learning (HFL) with weighted FedAvg. All control and weight-exchange channels are protected with NIST-standardized ASCON-128 authenticated encryption. Eight experimental variants are evaluated by crossing three binary axes (model architecture: MLP vs. CNN-1D; security: ASCON vs. plain JSON; topology: with vs. without a Fog leader/peer layer), yielding 58 federated runs consolidated under a unified site-reliability engineering (SRE) telemetry layer. The best configuration (CNN_ASCON_FOG) reaches 0.975 global accuracy with an ASCON-induced temporal overhead between 0.10 % and 0.27 % of the federated round and a ≈ 48 % per-message size overhead. Variants with authenticated encryption do not degrade convergence over the evaluated testbed, and the Fog topology improves accuracy when combined with ASCON but can hurt it without authentication. We discuss the validity boundaries of these findings, including the closed three-class label set, the heuristic gateway labeller, and the limited testbed scale.

Keywords: IoT Security · MQTT · TinyML · Hierarchical Federated Learning · ASCON-128 · Lightweight Cryptography · Edge-Fog-Cloud · Intrusion Detection System.

1 Introduction

The expansion of Internet of Things (IoT) and Industrial IoT (IIoT) deployments has enlarged the network attack surface of homes, laboratories, and industrial sites that increasingly rely on lightweight publish/subscribe protocols such as MQTT [15,2]. Detecting intrusions in those environments is challenging on at least three independent dimensions: (i) edge devices have severe memory and

energy constraints; (ii) centralising traffic captures into a single server is incompatible with the privacy and bandwidth profile of many deployments; and (iii) the federated training loop itself — model updates, control messages, deployment events — becomes a sensitive asset that must be protected in transit.

Existing solutions tend to address these dimensions *separately*: a substantial body of work studies TinyML-based intrusion detection on microcontrollers [3,7], a parallel line studies federated learning (FL) for IoT security [8,5], and a third one analyses lightweight authenticated encryption such as ASCON [13,9]. Few works measure all three together on physical hardware. The recently selected NIST ASCON standard [13] provides a 128-bit authenticated encryption-with-associated-data (AEAD) primitive specifically designed for constrained devices, but its empirical cost inside a real federated cycle on IoT MCUs is rarely reported end-to-end.

This paper addresses that integration gap and makes three contributions:

1. **An Edge–Fog–Cloud IDS architecture** for IoT/MQTT networks, built on ESP32-S3 nodes, Raspberry Pi 4 gateways and a PC coordinator, with two interchangeable topologies (a standard Edge→Fog→Cloud chain and a Fog *leader/peer* pre-aggregation variant).
2. **A lightweight federated TinyML pipeline** where only the last two dense layers of the on-device classifier (163 parameters, 652 bytes) are exchanged through weighted FedAvg, exported as embeddable C++ arrays for on-board inference without dynamic memory allocation.
3. **An ASCON-128 protected federated cycle** with measured latency and payload overhead against a functionally equivalent PLAIN baseline, evaluated jointly on the eight combinations of model, security and topology over 58 federated runs.

We do *not* claim that ASCON improves classification accuracy or that the system solves IoT security in general. The empirical contribution is the measurement of viability and overhead of the joint stack under a clearly stated threat model.

The rest of the paper is organised as follows. Section 2 positions the work in the literature. Section 3 describes the Edge–Fog–Cloud architecture. Section 4 details the federated TinyML pipeline. Section 5 covers the ASCON-128 security layer. Section 6 presents the experimental setup, and Section 7 reports the results. Section 8 discusses threats to validity, and Section 9 concludes and outlines future work.

2 Related Work

TinyML for IDS on MCUs. Dini *et al.* [3] survey AI-model toolchains (TensorFlow Lite Micro, STM32Cube.AI, Vitis AI) and show that successful integration on a microcontroller depends as much on the hardware/software stack as on the model itself. Tekin *et al.* [17] insist that on-device IDS work must report memory, latency and energy together with classification accuracy; Heydari and

Mahmoud [7] confirm this on smart-home hardware. Most reported pipelines still rely on capable servers and do not demonstrate deployment on real MCUs.

Federated learning for IoT security. Federated approaches reduce the need to centralise raw traffic, but introduce non-IID data, partial participation and communication overhead [8,1,16]. Hajj *et al.* [6] demonstrate a cross-layer federated IDS for IoT, while Ficco *et al.* [5] validate FL with transfer learning for on-board training on ESP32 and Raspberry Pi class devices. Tran *et al.* [18] quantify a memory gap of up to three orders of magnitude between Byzantine-robust aggregators (Krum, FLTrust, Fed-NGA) and typical MCU SRAM, arguing for hierarchical decomposition. Our work adopts that hierarchical view but focuses on confidentiality and integrity rather than Byzantine robustness.

Lightweight cryptography in the federated cycle. The NIST 800-232 standard formalises the ASCON family [13]; Kaur *et al.* [9] provide a comprehensive survey of ASCON implementations, attacks and countermeasures. In contrast, Mirzaali *et al.* [11] show experimentally on three Raspberry Pi devices that ad-hoc ultra-lightweight permutations used for authentication can be broken by disclosure and desynchronisation attacks, even when the protocol structure looks sound. These two references jointly motivate adopting a standardised AEAD inside the federated cycle and treating side-channel hardening as future work.

Gap. Existing work covers each of these three axes separately. We are not aware of a study that measures, on the same physical IoT testbed, the joint behaviour of TinyML inference, hierarchical federated training and authenticated lightweight encryption, with explicit comparison against a functionally equivalent plaintext baseline. This paper fills that gap.

3 System Architecture

The system is organised in three hierarchical layers (Fig. 1) that map onto the Edge, Fog and Cloud roles common in IoT references [19,18]. Each layer has a single responsibility, and the entire control and data plane is defined in terms of MQTT and HTTP exchanges, so the layers can be replaced independently.

Edge layer. Two ESP32-S3 microcontrollers per gateway publish features to the local Mosquitto broker via the `f1/features` topic. Each node executes a 1,139-parameter MLP classifier (≈ 4.5 kB) on thirteen flow features (packet counts, mean packet length, inter-arrival statistics, TCP flags) and reports per-flow predictions and feature vectors at a fixed cadence. To exercise heterogeneity, one node emits exclusively *normal* traffic, while a second node emits a calibrated mixture (40 % *normal*, 30 % *mqtt_bruteforce*, 30 % *scan_A*). No on-device training is performed on the MCU: the ESP32-S3 runs only inference and ASCON-128 in native C++.

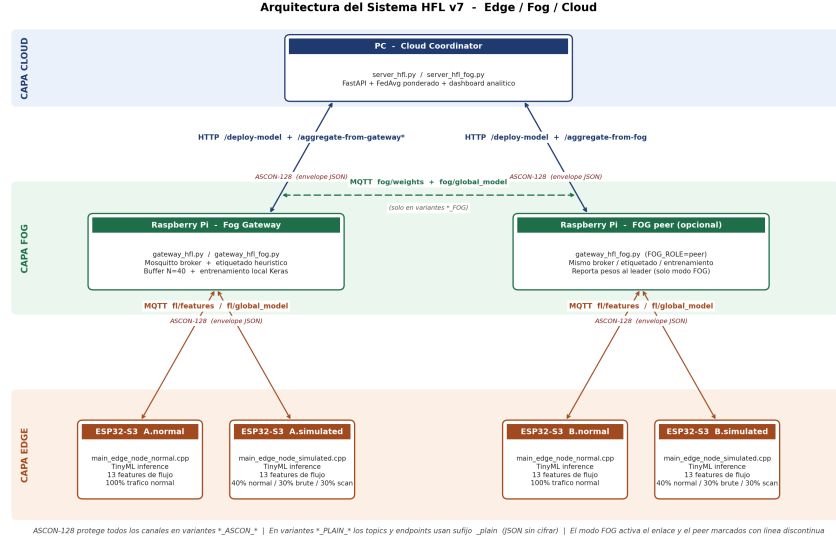


Fig. 1: Edge-Fog-Cloud architecture of the proposed IDS. The dashed links are active only in the Fog (leader/peer) topology variant; all channels are protected with ASCON-128 in the *_ASCON_* variants.

Fog layer. A Raspberry Pi 4 acts simultaneously as the Mosquitto broker, the local trainer and an HTTP bridge to the coordinator. Incoming feature vectors are labelled by a heuristic rule (Section 4), accumulated into a buffer of $N_s = 30$ samples, used to train the federated layers of the classifier for $E = 5$ local epochs, and finally serialised together with the local accuracy and sample count for transmission to the PC server. In the Fog topology variant a second Raspberry Pi participates as a *peer*: it ships its local weights to the *leader* through the `fog/weights` topic, the leader applies an intermediate FedAvg and sends the pre-aggregated update to the cloud, then redistributes the received global model through `fog/global_model`.

Cloud layer. A FastAPI server (`server_hfl.py` or `server_hfl_fog.py`) collects updates from $K = 2$ gateways, performs a weighted FedAvg, and publishes the new global model back to the gateways through HTTP (`/deploy-model`). It also acts as the sink for the SRE telemetry described later. The server is assumed to be inside the trusted computing base and is the only point that observes weights from *all* gateways.

Threat model. We assume a passive or active network attacker that can eavesdrop or modify wireless traffic between the ESP32 nodes, the Raspberry Pi gateways and the PC coordinator, but cannot compromise the endpoints. ASCON-128 over every channel addresses confidentiality and integrity of features, weights and

Table 1: Parameter distribution of the on-device MLP classifier. Federated layers (in *italics*) are the only ones exchanged each round.

Layer	Input	Output	Parameters
Dense 1 (ReLU)	13	32	$13 \times 32 + 32 = 448$
Dense 2 (ReLU)	32	16	$32 \times 16 + 16 = 528$
<i>Dense 3 (ReLU, fed.)</i>	16	8	$16 \times 8 + 8 = 136$
<i>Dense 4 (Softmax, fed.)</i>	8	3	$8 \times 3 + 3 = 27$
Total			1,139

control messages; nonces are constructed from a 32-bit truncated timestamp and a monotonic counter to detect replay. Out of scope for this paper are Byzantine clients and data poisoning, physical side-channel attacks against ASCON, and membership inference over the federated weights; these limitations are discussed explicitly in Section 8.

4 Federated TinyML Pipeline

4.1 On-device classifier

The on-device classifier is a fully connected network with $13 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 3$ topology (Table 1), totalling 1,139 parameters in 32-bit floating point (≈ 4.5 kB), which fits comfortably in the ESP32-S3 SRAM without quantisation. The federated cycle exchanges only the last two dense layers (**w3**, **b3**, **w4**, **b4**, a total of 163 parameters or 652 bytes per round); the lower layers are trained offline and frozen on the device. A second branch replaces the dense lower block with a 1D-CNN (**Conv1D(32)** $\rightarrow$ **Conv1D(16)** $\rightarrow$ GAP) whose convolutional and Batch-Norm layers are fused and kept fixed on the MCU, leaving the same federated dense block as the only trainable head. Federating only the final layers follows the FL+TL strategy reported in [5,16] and bounds the payload size irrespective of the underlying architecture.

4.2 Hierarchical FedAvg protocol

Each gateway accumulates $N_s = 30$ labelled samples per round and trains locally for $E = 5$ epochs. The PC server then applies weighted FedAvg:

$$\mathbf{W}_{\text{global}}^{(t+1)} = \sum_{k=1}^K \frac{n_k}{\sum_{j=1}^K n_j} \mathbf{W}_k^{(t)}, \quad (1)$$

where n_k is the number of samples used in gateway k 's local update and $\mathbf{W}_k^{(t)}$ are its federated weights at round t . In the Fog variant Eq. 1 is first applied locally at the leader on the messages received from its peers, and the resulting pre-aggregated update is sent to the PC server.

Labels are assigned at the gateway by a heuristic calibrated on the distribution of the public datasets: `num_pkts` ≥ 50 and `num_psh` ≥ 10 produces the *bruteforce* label; `num_pkts` ≤ 5 with small `mean_pkt_len` and at least one PSH produces *scan_A*; otherwise the sample is mapped to *normal* if the byte counts fall in the expected range, and discarded otherwise. This heuristic is a known source of label noise and is treated explicitly as a validity threat (Section 8).

4.3 Datasets and classes

The system targets three traffic classes (*normal*, *mqtt_bruteforce*, *scan_A*) selected because they are representative of what a single MQTT-enabled node may observe in a smart home or laboratory setting. The classes are derived from public IoT/IDS corpora such as MQTT-IoT-IDS2020 and TON_IoT [12,4,14] after unifying schemas and converting captures into the 13-dimensional flow feature space described above. A central training set with 256,276 flows is reserved for the centralised classical baselines reported in Section 7.

5 ASCON-128 Security Layer

ASCON-128 provides authenticated encryption with associated data (AEAD) using a 128-bit pre-shared symmetric key, a 128-bit nonce constructed from a truncated timestamp and a monotonic counter, and a 128-bit authentication tag. We chose the original ASCON-128 specification (64-bit rate) over ASCON-128a (128-bit rate) because the payload sizes in the system (13 floats for features and ~ 3.6 kB for weight messages) are small enough that the throughput difference is dominated by other factors, while the higher number of permutation rounds in ASCON-128 preserves the security margin recently revisited by Kaur *et al.* [9].

We protect the four (or, in the Fog variant, six) channels of the system, shown in Table 2. Both implementations — the native C++ header used on the ESP32-S3 and the Python implementation used on Raspberry Pi and PC — share test vectors so interoperability is guaranteed across architectures. The encrypted payload is encapsulated in a JSON envelope with three base64-encoded fields: `{"ct": "...", "tag": "...", "nonce": "..."}`, exactly as in [13]. All encryption and decryption operations are instrumented with high-resolution timers (`micros()` on the MCU, `time.perf_counter()` in Python) and persisted in a CSV log together with the federated round identifier and payload sizes, enabling the overhead breakdown reported in Section 7.

A PLAIN baseline branch (`hf1_v7-no-ascon`) is kept synchronised with the secure branch and ships the same JSON payloads without encryption, so that overhead in time and bytes can be attributed solely to the cryptographic layer.

6 Experimental Setup

The testbed consists of four ESP32-S3 microcontrollers (two per gateway), two Raspberry Pi 4 (4 GB) acting as gateways and, in the Fog variant, as leader/peer,

Table 2: Communication channels protected by ASCON-128.

Channel	Protocol	Content	Implementation
ESP32 → RPi	MQTT	Feature vectors (13 floats)	C++
RPi → PC	HTTP	Local weights + metrics	Python
PC → RPi	HTTP	Global model	Python
RPi → ESP32	MQTT	Global model	Python → C++
<i>Fog topology additionally enables:</i>			
peer → leader	MQTT	Local weights (pre-agg.)	Python
leader → peer	MQTT	Global model (re-dist.)	Python

 Table 3: Experimental variants. Cells indicate the value of each design axis; *Purpose* states the comparison each variant enables.

ID	Edge model	Security	Topology	Purpose
E1	MLP (RN)	ASCON-128	NoFOG	Reference integral system
E2	MLP (RN)	PLAIN	NoFOG	Isolate ASCON cost in RN
E3	MLP (RN)	ASCON-128	FOG	Fog with authenticated RN
E4	MLP (RN)	PLAIN	FOG	Fog without authentication, RN
E5	CNN-1D	ASCON-128	NoFOG	Effect of model change
E6	CNN-1D	PLAIN	NoFOG	Isolate ASCON cost in CNN
E7	CNN-1D	ASCON-128	FOG	Best-case configuration
E8	CNN-1D	PLAIN	FOG	Fog without authentication, CNN

and a single PC running the coordinator and SRE telemetry sink. All devices share a private 2.4GHz Wi-Fi network. The same hardware is reused across all variants so observed differences can be attributed to a single design axis at a time.

We instantiate the full $2^3 = 8$ factorial design across three binary axes: model architecture (RN $\equiv$ MLP vs. CNN-1D), security mode (ASCON-128 vs. PLAIN), and topology (NoFOG vs. FOG), summarised in Table 3. Each variant is run at least 20 federated rounds and replicated up to 8 times. The unified telemetry layer (*Análisis de Modelos/Completo/*) records per-round accuracy and loss, per-channel transport latency at the p95 percentile, per-message plaintext and ciphertext sizes, and a canonical event stream, producing 58 runs in total after consolidation.

Performance is evaluated along four orthogonal axes: classification quality (accuracy, F1 macro and weighted, precision, recall, MCC), cryptographic overhead (encryption/decryption time, byte and percentage increase per message, fraction of the federated round), federated efficiency (rounds to convergence, accuracy per round, communication cost per round), and edge viability (model footprint and inference latency on the ESP32-S3).

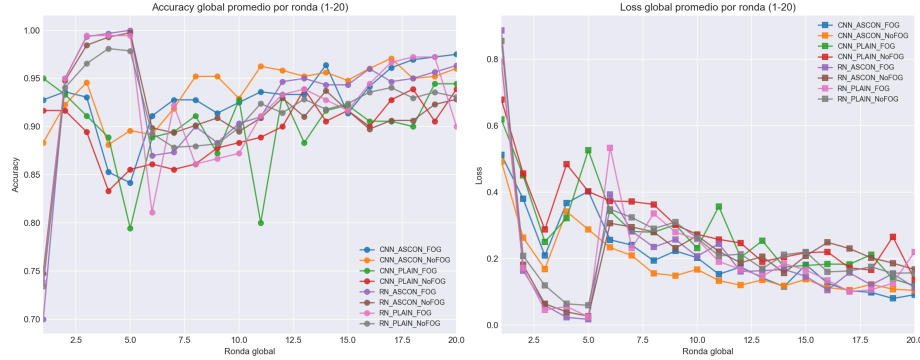


Fig. 2: Global accuracy (left) and global loss (right) averaged across attempts for each variant, rounds 1–20. All variants cross 0.90 accuracy between rounds 5 and 10; CNN variants stabilise at a higher plateau.

7 Results and Discussion

7.1 Centralised baselines and lightweight model comparison

Classical centralised models trained on the full 256,276-flow corpus (80/20 stratified split, 5-fold cross-validated hyperparameter search) establish the achievable ceiling. Random Forest reaches an accuracy of 0.9994 with an F1-weighted of 0.9994 and an MCC of 0.9989, followed by Decision Tree (0.9987) and SVM-RBF (0.9968); linear models drop by 5–6 points, confirming that the decision boundaries between the three classes are not linearly separable in the 13-feature space. Random Forest and SVM-RBF, however, exceed the 512 kB SRAM of the ESP32-S3 by orders of magnitude and are not deployable as on-device classifiers.

An offline comparison of MCU-compatible architectures over the same features (MLP, CNN-1D, Residual-MLP, Tiny Transformer) places the three viable architectures within ± 0.6 percentage points of 0.905 accuracy. The MLP is retained as the primary on-device classifier for its smaller inference cost, while the CNN-1D is preserved as a second iteration to study whether higher expressive capacity translates into a real advantage inside the federated loop.

7.2 Federated convergence

Fig. 2 shows the per-round trajectory of accuracy and loss for the eight variants over the first twenty rounds. Convergence is reached between rounds 5 and 10 across all variants; the loss drops from ≈ 0.8 to the 0.1–0.2 range in the same window. The CNN variants, in particular CNN_ASCON_FOG and CNN_ASCON_NoFOG, exhibit a more expressive weight trajectory (Fig. 3): `w3_mag` and the class-associated rows `w4_brute` and `w4_scan` grow steadily, while the RN variants stabilise early around `w3_mag` ≈ 0.29 . No variant diverges in the first twenty rounds; the noisiest behaviour appears in `w4_normal` of CNN_PLAIN_NoFOG, which

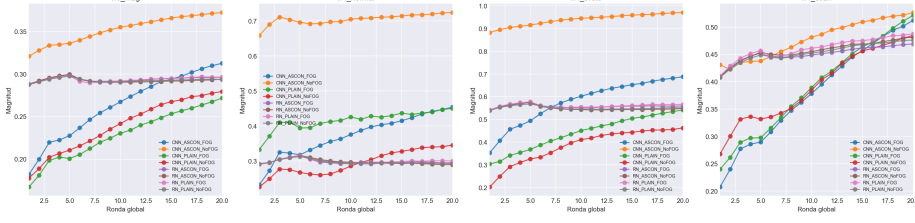


Fig 3: Mean magnitude of the four federated weight tensors per variant across rounds 1–20.

Table 4: Summary of the eight HFL variants. *GW p95*: p95 decryption latency at the gateway (ESP32 $\rightarrow$ RPi). *Srv p95*: p95 decryption latency at the server (RPi $\rightarrow$ PC). *Overhead*: mean bytes added by the ASCON+Base64 envelope (zero for PLAIN by construction).

Variant	Acc.	Loss	Round (s)	GW p95	Srv p95	Ovh. (B)
CNN ASCON FOG	0.9750	0.091	58.6	4.53	37.5	1098
CNN ASCON NoFOG	0.9604	0.104	59.0	4.55	24.5	1049
CNN PLAIN FOG	0.9444	0.119	54.9	0.16	50.5	—
CNN PLAIN NoFOG	0.9389	0.136	52.0	0.15	0.0	—
RN ASCON FOG	0.9633	0.112	71.1	4.37	50.5	1084
RN ASCON NoFOG	0.9381	0.154	72.9	4.65	26.7	1050
RN PLAIN FOG	0.9000	0.220	67.1	0.16	0.0	—
RN PLAIN NoFOG	0.9298	0.158	51.0	0.17	0.0	—
Mean (8)	0.9437	0.138	61.3	2.34	17.4	636

is consistent with the absence of both Fog aggregation and authenticated tag verification.

7.3 Comparison across variants

Table 4 summarises the eight variants. The best configuration is `CNN_ASCON_FOG` with a final accuracy of 0.9750 and final loss of 0.0910, beating its `PLAIN` counterpart by 3.0 points and its `NoFOG` counterpart by 1.5 points. The most striking pairwise comparison appears in the `RN` family: `RN_ASCON_FOG` reaches 0.9633 against 0.9000 for `RN_PLAIN_FOG`, a 6.3-point gap. This is consistent with the hypothesis that authenticated-tag verification filters out corrupted or duplicate messages before they enter the local training buffer; however, our experiments do not include controlled fault injection, so we report this only as an observation and not a causal claim.

Disaggregated by class, the best variant exhibits a macro-F1 of ≈ 0.972 and weighted-F1 of ≈ 0.975 , with per-class recall above 0.96 in all three classes (slightly lower for `scan_A` where short burst flows partially overlap statistically with low-

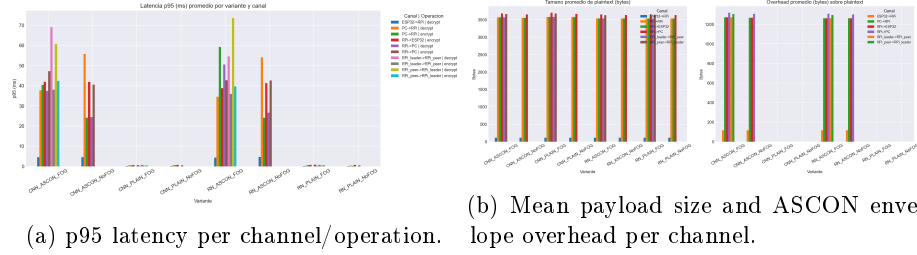


Fig. 4: Per-channel cryptographic and communication overhead. PLAIN variants report only JSON serialisation time (sub-millisecond) and zero envelope overhead by construction.

Table 5: Size overhead per ASCON variant. PLAIN omitted (zero by construction).

Variant	Plaintext (B)	Overhead (B)	Ratio (%)
CNN ASCON FOG	2365	1098	46.5
CNN ASCON NoFOG	2143	1049	48.9
RN ASCON FOG	2323	1084	46.6
RN ASCON NoFOG	2162	1050	48.5
Mean	2248	1070	47.6

intensity *normal* traffic). The MCC value is ≈ 0.96 , confirming that the global accuracy is not inflated by the majority class.

7.4 Cryptographic overhead

ASCON-128 decryption at the ESP32-S3 gateway entry takes 4.4–4.7 ms p95, essentially independent of model and topology. At the PC server the Python implementation reports 24–27 ms p95 in NoFOG and 37–50 ms p95 in FOG (the leader re-encrypts pre-aggregated updates, which doubles the cryptographic operations on the upstream path). Summed over a federated round, ASCON contributes between 73 ms (NoFOG) and 140 ms (FOG); over rounds of 51–73 s this is between 0.10 % and 0.27 % of the round time (Fig. 4, left).

In terms of payload, the ASCON envelope adds on average 1.07 kB per message, or ≈ 48 % relative to a 2.25 kB plaintext payload (Table 5). This breaks down as a fixed 16-byte tag, 16-byte nonce, a $\times 4/3$ base64 expansion, and a ~ 30 -byte JSON envelope. On a standard 54 Mbps Wi-Fi link the absolute additional transmission time per round (3.2 kB in NoFOG, 5–6 kB in FOG) is below one millisecond and therefore negligible.

7.5 Federated vs. centralised

Compared to the Random Forest centralised ceiling (0.9994), the best federated variant reaches 0.9750, a 2.4-point gap. The MLP variants pay a larger gap (5–6 points), consistent with the FL non-IID literature [8,1]. In exchange, the federated system exchanges only 163 parameters (652 bytes) per round between each gateway and the coordinator, compared with the millions of feature values that a centralised collector would have to ingest — a reduction in transferred information of more than 99.99 %.

8 Threats to Validity

We make the following limitations explicit, because they constrain the generalisability of the reported numbers.

Closed three-class label set. The system is a closed-set classifier over *normal*, *mqtt_bruteforce* and *scan_A*. A traffic vector outside this set (e.g. *slowloris*, MQTT broker spoofing, application-layer DoS, covert exfiltration over QoS-2) will be mapped to the closest known class by softmax similarity, with two failure modes: a silent false negative if it resembles *normal*, or a noisy false positive if it resembles a known attack class. Open-set recognition or an explicit anomaly-detection head is required to mitigate this and is left as future work.

Heuristic gateway labelling. Labels at the gateway are assigned by fixed thresholds on `num_pkts`, `mean_pkt_len` and `num_psh` calibrated on the public datasets. This produces label noise relative to the ground-truth annotations and partially accounts for the residual gap versus the centralised baseline. A learned labelling head or a fully ground-truthed real capture would tighten this bound.

Limited testbed scale. The reported numbers come from two gateways and four edge nodes, sufficient to validate the cycle but not large enough to characterise behaviour at the scale of dozens or hundreds of gateways. At larger scale the PC coordinator would become a bottleneck and a genuinely distributed Fog plane (multiple leaders or sharded topics) would be required. Stronger non-IID distributions between gateways are also expected to widen the gap to the centralised baseline.

Simulated traffic, no energy reporting. The ESP32-S3 nodes emit calibrated traffic rather than capturing real MQTT flows. Energy consumption was not measured in the present setup; both points are flagged in the future-work section.

Cryptographic threat model and key management. ASCON-128 is evaluated as a black-box AEAD. We did not perform side-channel attacks (power analysis, EM, fault injection) against the implementation, and the system assumes deployment in a physically controlled environment. A 128-bit symmetric key is pre-shared across all devices to simplify the prototype; a production deployment

would require key rotation, secure onboarding of new nodes and revocation of compromised devices, none of which are addressed here. Mirzaali *et al.* [11] provide a methodological template for an active attack campaign on Raspberry Pi that we intend to replicate as future work.

Statistical significance. Per-variant means are computed over 5–8 attempts. Variance and confidence intervals are reported in the underlying telemetry CSVs but not in the summary table for space reasons; absolute differences below approximately 1.5 percentage points should not be over-interpreted.

9 Conclusions and Future Work

We presented an Edge–Fog–Cloud intrusion-detection system for IoT/MQTT networks that jointly evaluates TinyML inference on ESP32-S3, hierarchical federated learning across Raspberry Pi 4 gateways, and ASCON-128 authenticated encryption over the full federated cycle. Eight configurations crossing model, security and topology were measured under a unified telemetry layer over 58 federated runs. The best variant, CNN_ASCON_FOG, reaches 0.975 global accuracy with an ASCON-induced temporal overhead below 0.3 % of the round time and an envelope size overhead of ≈ 48 % per message; the on-device model footprint of ≈ 4.5 kB leaves more than 60 % of the ESP32-S3 SRAM free during operation. The Fog topology improves accuracy when combined with ASCON but can degrade it without authentication, suggesting that pre-aggregation is best deployed in conjunction with end-to-end integrity guarantees.

The contribution should be read as a functional end-to-end proof of the integration, not as a production-ready solution: the testbed is small, the label set is closed, and the cryptographic deployment relies on a pre-shared key. Three directions stand out as immediate future work. The first is Byzantine-robust aggregation and stronger non-IID handling — FedProx, SCAFFOLD or hierarchical schemes such as EdgeTrust-Shard [10,18] — to address the gap that real deployments are likely to exhibit. The second is moving from the pre-shared key used in this prototype to a realistic key-management plane with rotation, secure onboarding and revocation, complemented by a side-channel hardening campaign of the ASCON implementation along the lines surveyed by Kaur *et al.* [9] and an active attack replication in the spirit of Mirzaali *et al.* [11]. The third is widening the label set beyond the three classes evaluated here (e.g. MQTT flooding, bot-net traffic, credential abuse) and pairing the closed-set classifier with an open-set or anomaly detection head, so that traffic outside the trained catalogue is flagged as unknown rather than mapped to the nearest known class.

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